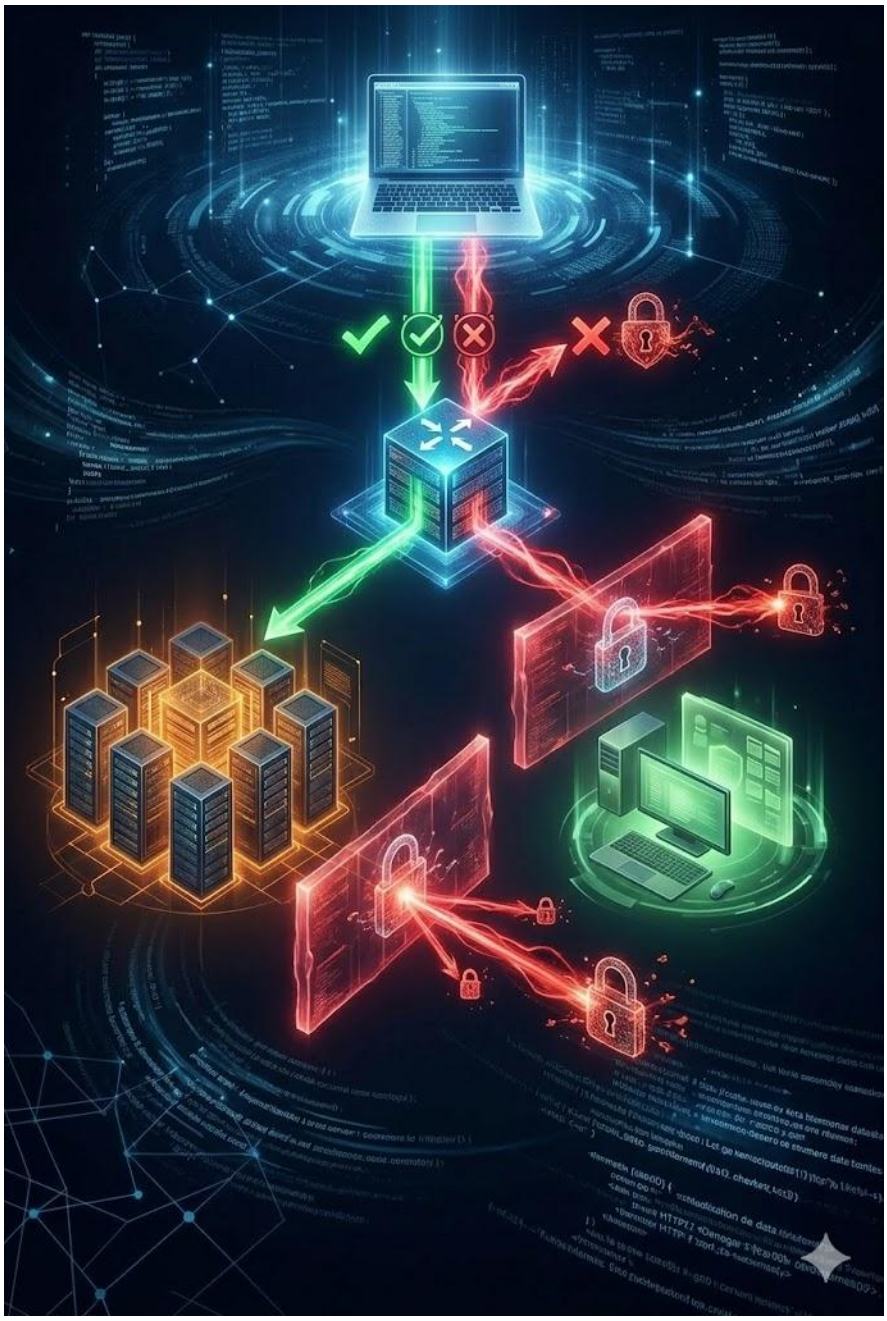


CONFIGURACIÓN DE DMZ

CON CISCO PACKET TRACER

Jesús Álvarez | Ciberseguridad | 19.03.2026

OBJETIVO DEL LABORATORIO

El propósito fundamental de esta práctica es el diseño, configuración y aseguramiento de una infraestructura de red que utiliza una Zona Desmilitarizada (DMZ). Los objetivos principales alcanzados son:

- **Implementación de NAT Estático**: Configurar la traducción de direcciones necesaria para que un servidor con IP privada sea accesible desde una red pública (Internet) de forma controlada.
- **Segmentación y Aislamiento:** Dividir la red en tres áreas de confianza distintas (LAN, DMZ y Externa) para evitar que un compromiso en el servidor web permita el acceso a los datos de la red interna.
- **Aplicación de Políticas de Seguridad (ACLs):** Diseñar y aplicar listas de control de acceso que filtren el tráfico basándose en protocolos y puertos específicos (como el puerto 80 para HTTP), aplicando el principio de "mínimo privilegio" para maximizar la seguridad perimetral**.**

TOPOLOGÍA IMPLEMENTADA

- **Cantidad de redes:** 3 (LAN Interna, DMZ, Red Externa/Internet).
- **Dispositivos usados:** 1 Router Cisco 2911 (Router_FW), 3 Switches 2960, 2 PCs y 1 Servidor.
- **Descripción de zonas:**
    - **LAN:** Red privada donde residen los usuarios internos (PC_Internal).
    - **DMZ:** Zona aislada que contiene servicios públicos (Server_DMZ) para que sean accesibles desde fuera sin entrar a la LAN.
    - **Externa:** Simula el tráfico proveniente de Internet (PC_External).

PLAN DE DIRECCIONAMIENTO IP

Dispositivo	IP	Mascara	Gateway
PC_Internal	192.168.1.10	255.255.255.0	192.168.1.1
Server_DMZ	192.168.2.10	255.255.255.0	192.168.2.1
PC_External	192.168.3.10	255.255.255.0	192.168.3.1
Router_FW Gi0/0 (LAN)	192.168.1.1	255.255.255.0	N/A
Router_FW Gi0/1 (DMZ)	192.168.2.1	255.255.255.0	N/A
Router_FW Gi0/2 (Ext)	192.168.3.1	255.255.255.0	N/A

CONFIGURACIÓN APLICADA (RESUMEN)

- **NAT Estático:** Se mapeó la IP privada del servidor a la IP pública del router: ip nat inside source static 192.168.2.10 192.168.3.1
- **ACLs de Seguridad:**

- **ACL 100 (WAN):** Deniega el acceso desde la DMZ hacia la red LAN interna para evitar saltos laterales en caso de ataque, permitiendo únicamente responder a conexiones que ya han sido establecidas desde adentro.
- **ACL 101 (DMZ):** Permite el tráfico web (HTTP en puerto 80 y HTTPS en puerto 443) y pruebas de conexión (ping) desde el exterior hacia el servidor de la DMZ
- **ACL 102 (LAN):** Aísla el tráfico de la oficina hacia la DMZ para cumplir con la política de seguridad estricta del laboratorio.

VERIFICACIONES REALIZADAS

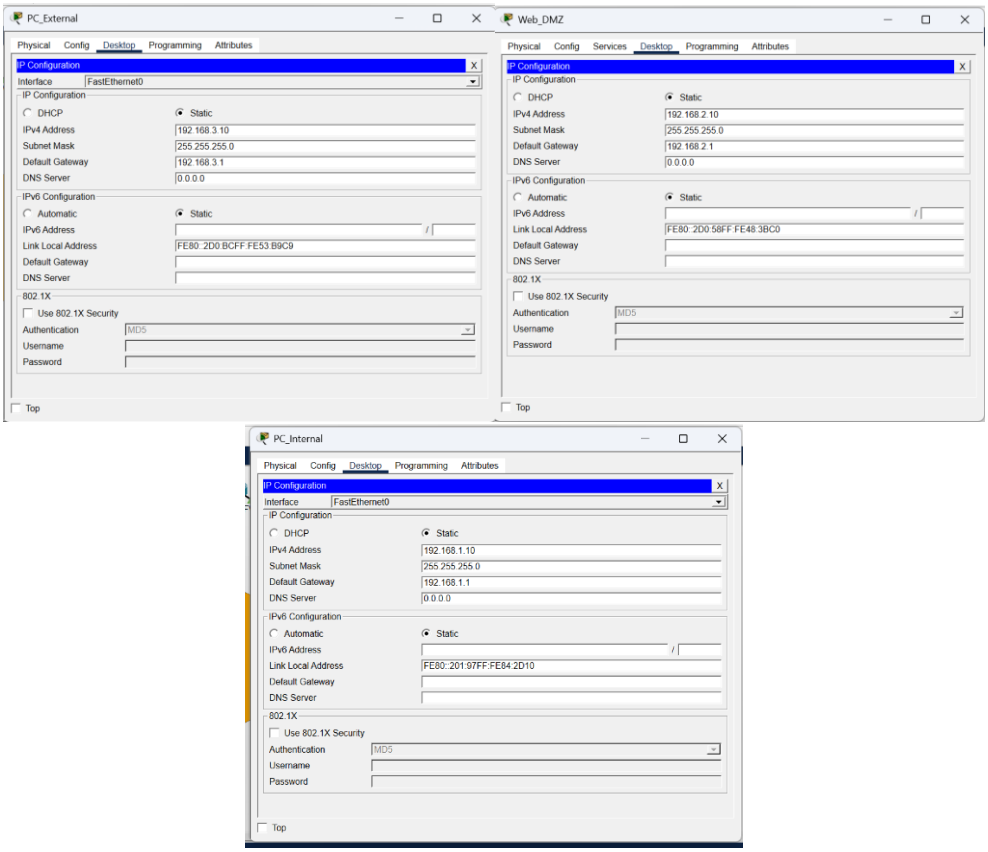
- Desde PC_External a (Web Browser): Exitoso
- Desde PC_External (command prompt): Exitoso
- Desde PC_Internal a web DMZ (Web Browser): Exitoso
- Desde server-PT a PC_Internal (command prompt): Bloqueado (ACL 100 protegiendo el servidor)

CONCLUSIÓN

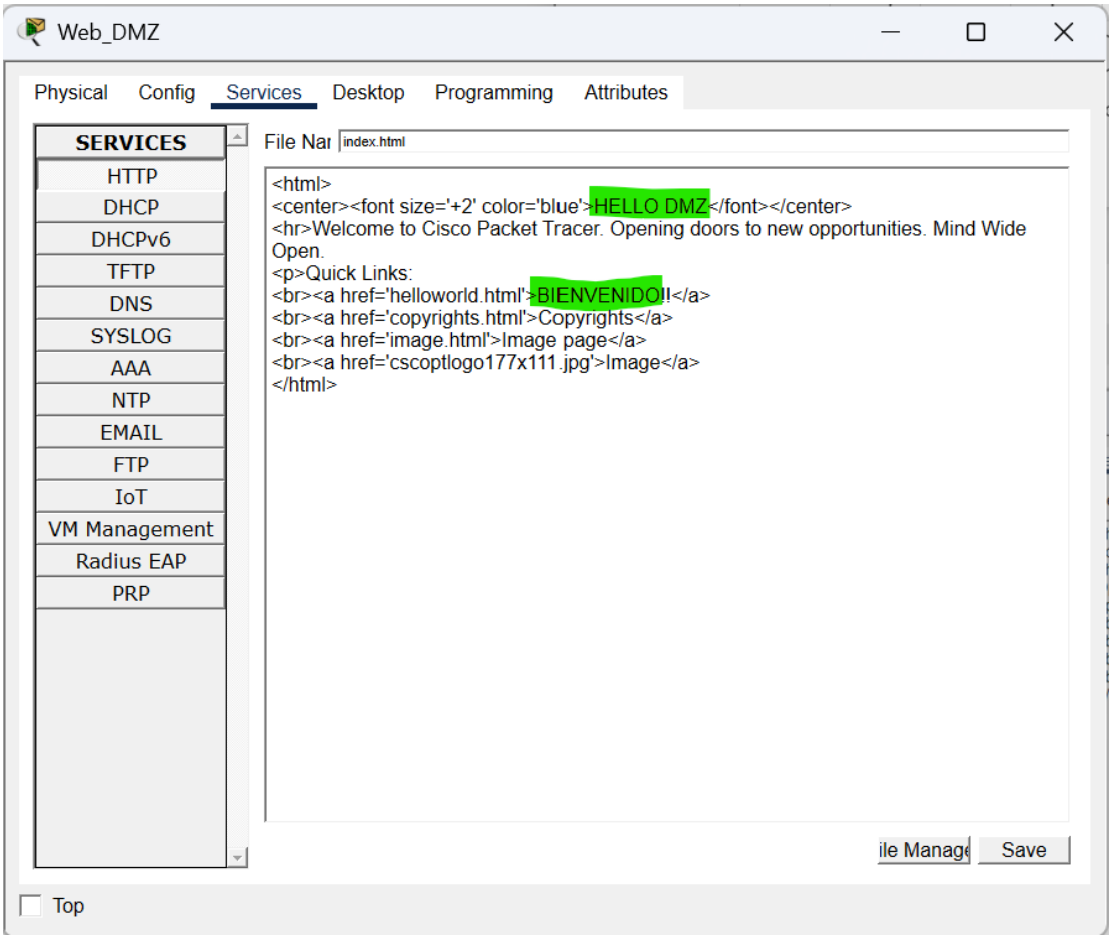
En este laboratorio aprendí que una DMZ no solo consiste en separar servidores, sino en gestionar mediante ACLs y NAT quién puede ver a quién. Es fundamental denegar por defecto y permitir solo lo necesario. Un detalle crítico fue asegurar que el tráfico de la DMZ nunca pueda iniciarse hacia la LAN interna.

CAPTURAS DE EVIDENCIAS

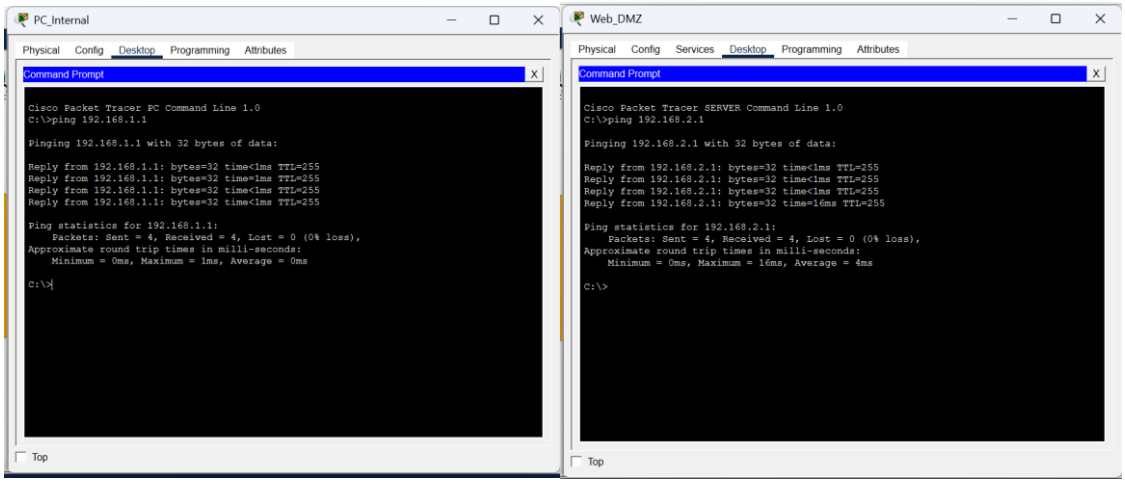
- Configuración de dispositivos:

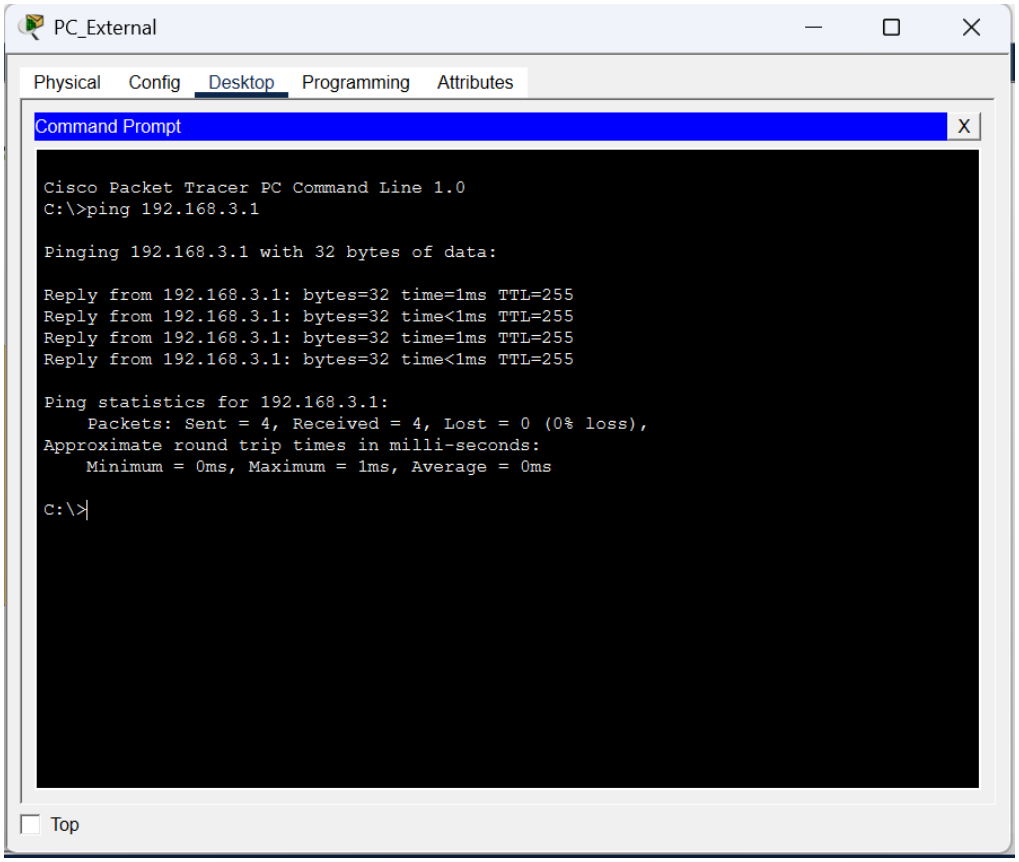


- Configuración interfaz del router:

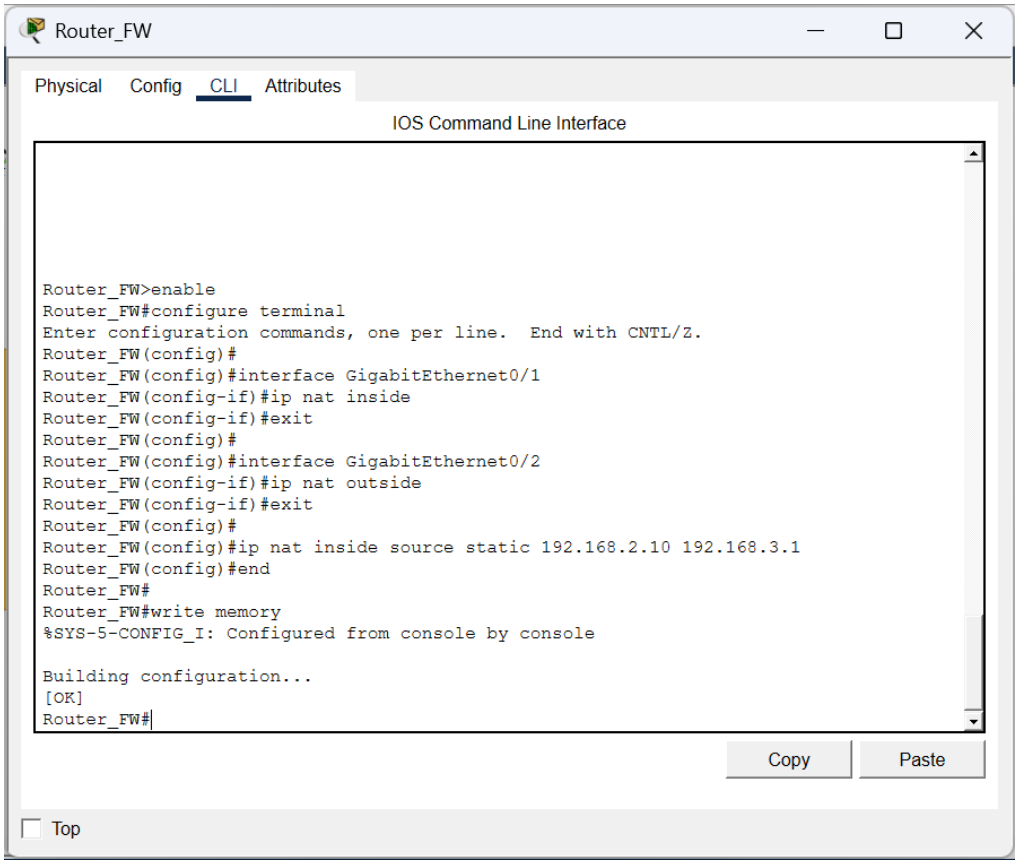


- Verificación de conectividad básica

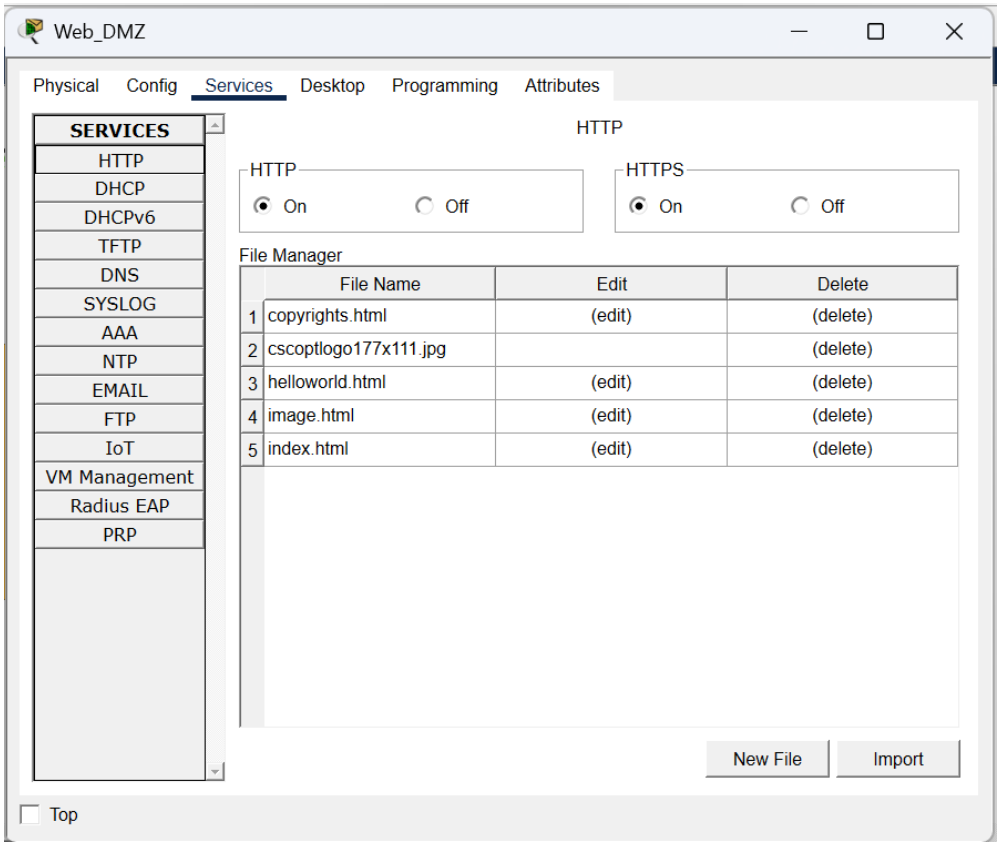


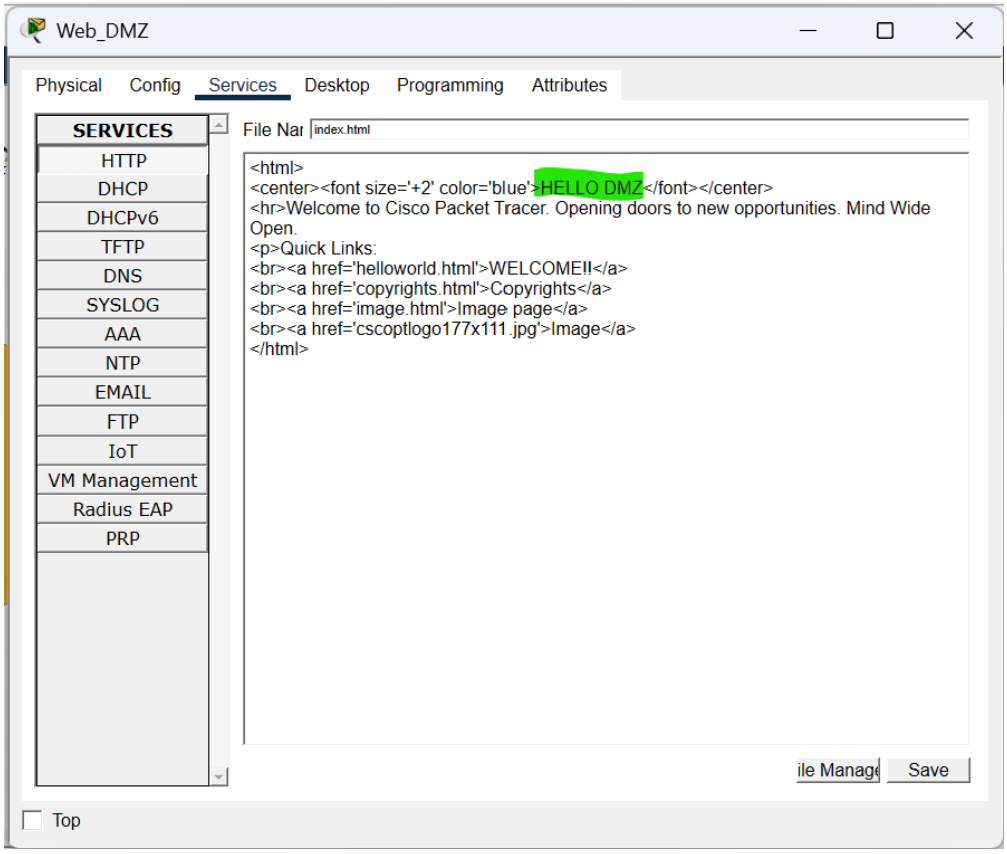


- Configuración de NAT Estático en el router

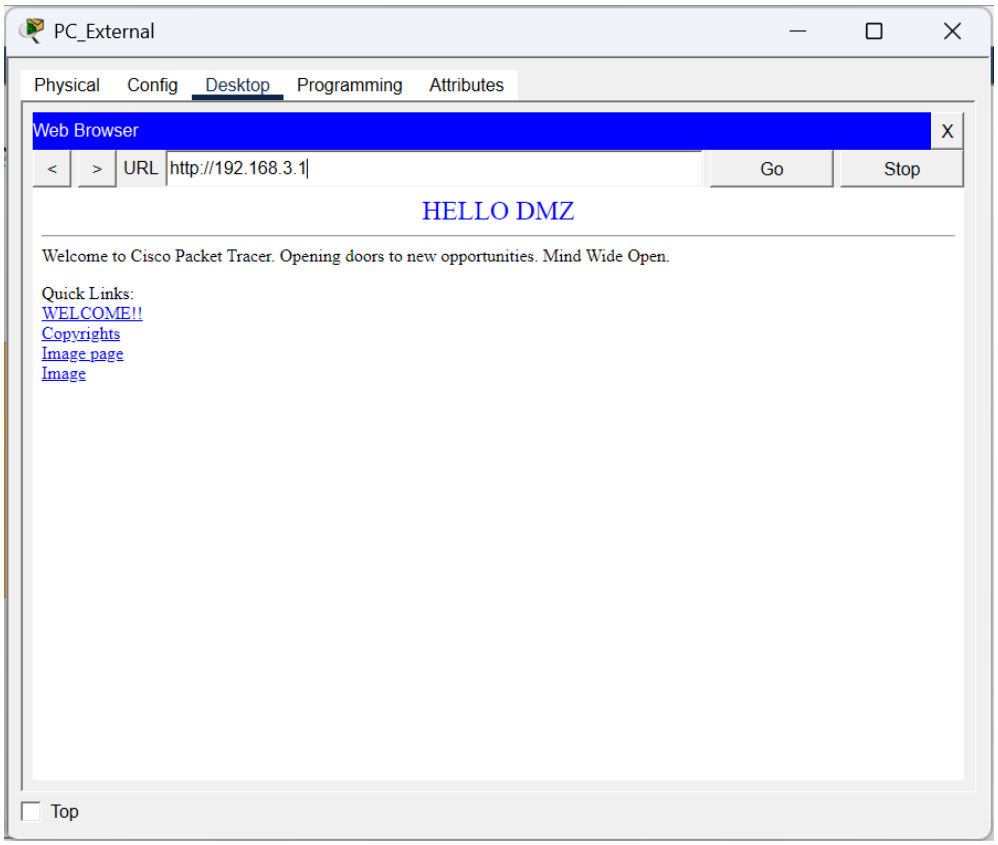


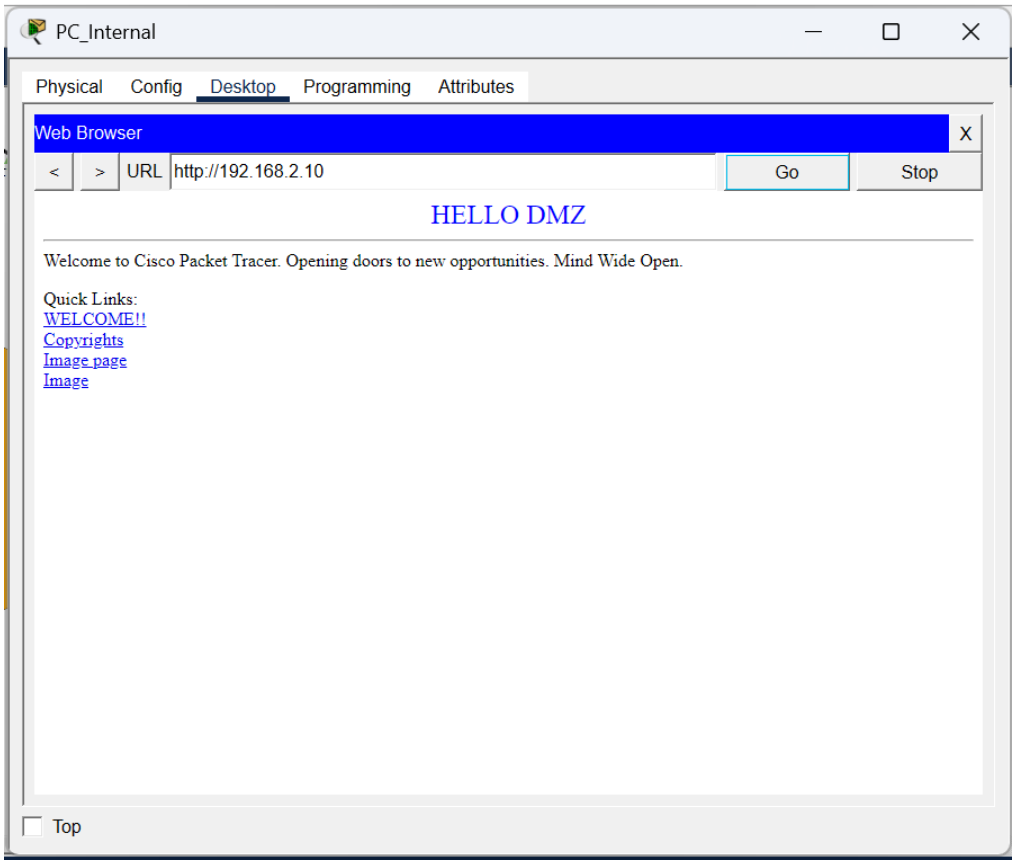
Activación de servicios Web en el servidor DMZ





Prueba de Acceso Web Inicial (ANTES de ACLs de Seguridad)





Configuración de ACLs para Seguridad

Router_FW

Physical Config CLI Attributes

IOS Command Line Interface

```
Router_FW(config)#ip nat inside source static 192.168.2.10 192.168.3.1
Router_FW(config)#end
Router_FW#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router_FW#enable
Router_FW#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router_FW(config)#no access-list 101
Router_FW(config)#ip access-list extended 101
Router_FW(config-ext-nacl)#10 permit icmp any any
Router_FW(config-ext-nacl)#20 permit tcp any host 192.168.3.1 eq 80
Router_FW(config-ext-nacl)#30 permit tcp any host 192.168.3.1 eq 443
Router_FW(config-ext-nacl)#exit
Router_FW(config)#ip access-list extended 100
Router_FW(config-ext-nacl)#10 permit icmp any 192.168.1.0 0.0.0.255 echo-reply
Router_FW(config-ext-nacl)#20 permit tcp any 192.168.1.0 0.0.0.255 established
Router_FW(config-ext-nacl)#30 permit tcp any 192.168.3.0 0.0.0.255 established
Router_FW(config-ext-nacl)#40 permit icmp any 192.168.3.0 0.0.0.255 echo-reply
Router_FW(config-ext-nacl)#50 deny ip 192.168.2.0 0.0.0.255 192.168.1.0
0.0.0.255
Router_FW(config-ext-nacl)#60 permit ip any any
Router_FW(config-ext-nacl)#exit
Router_FW(config)#interface GigabitEthernet0/1
Router_FW(config-if)#ip access-group 100 in
Router_FW(config-if)#exit
Router_FW(config)#interface GigabitEthernet0/2
Router_FW(config-if)#ip access-group 101 in
Router_FW(config-if)#exit
Router_FW(config)#end
Router_FW#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router_FW#
```

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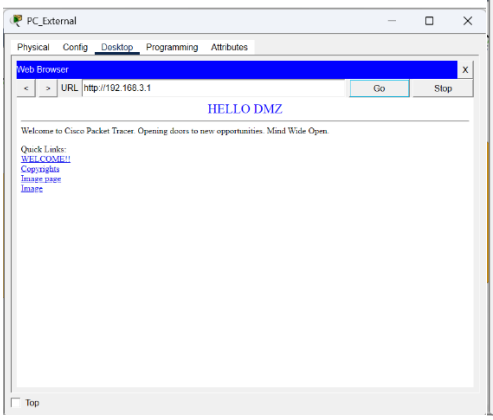
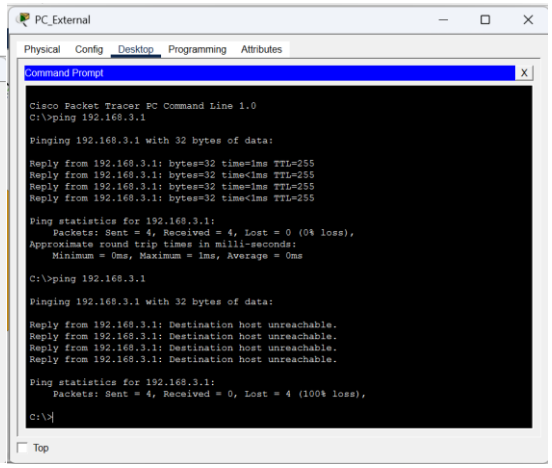
Top

Verificación Final de Seguridad y Funcionalidad


PC_External
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
http://192.168.3.1
Go
Stop
HELLO DMZ
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
WELCOME!!
Copyrights
Image page
Image
Top


PC_External
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.1
Pinging 192.168.3.1 with 32 bytes of data:
Reply from 192.168.3.1: bytes=32 time=1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time=1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.3.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.3.1
Pinging 192.168.3.1 with 32 bytes of data:
Reply from 192.168.3.1: Destination host unreachable.
Reply from 192.168.3.1: Destination host unreachable.
Reply from 192.168.3.1: Destination host unreachable.
Reply from 192.168.3.1: Destination host unreachable.
Ping statistics for 192.168.3.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
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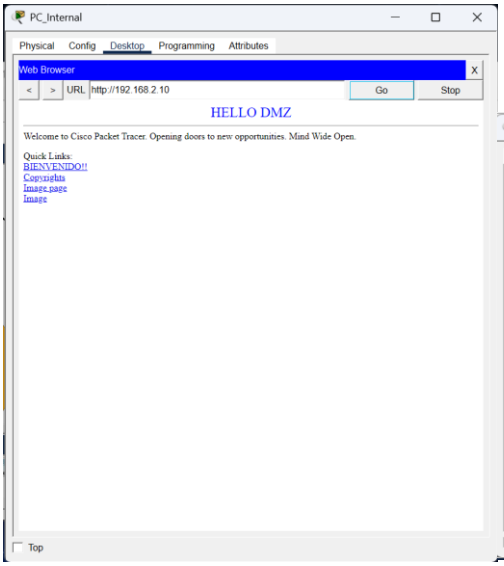
PC_Internal
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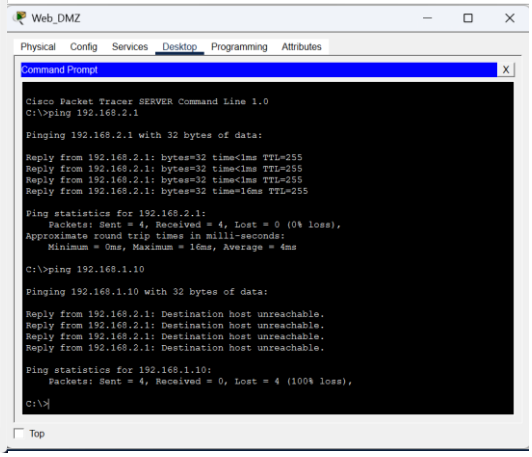
Web_DMZ
Physical
Config
Services
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time=16ms TTL=255
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 16ms, Average = 4ms
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
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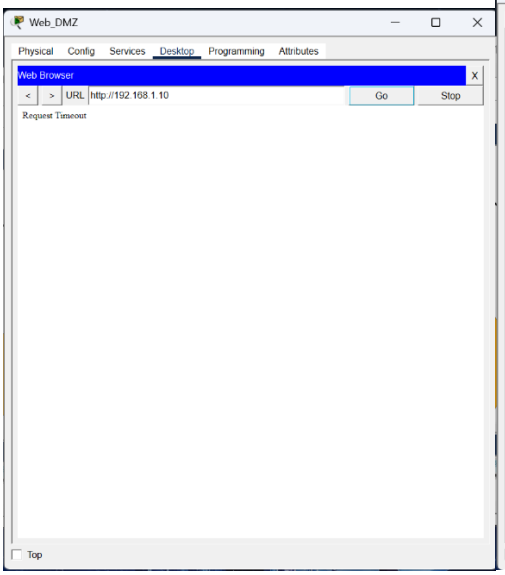
Web_DMZ
Physical
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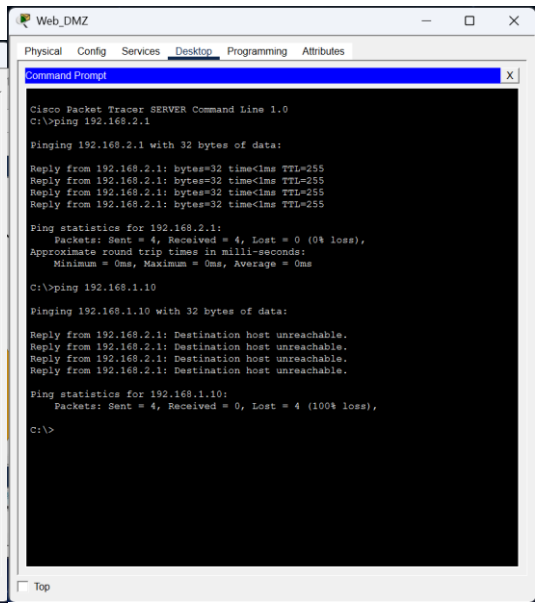
Services
Desktop
Programming
Attributes
Web Browser
URL
http://192.168.1.10
Go
Stop
Request Timeout
Top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Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.2.1
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Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
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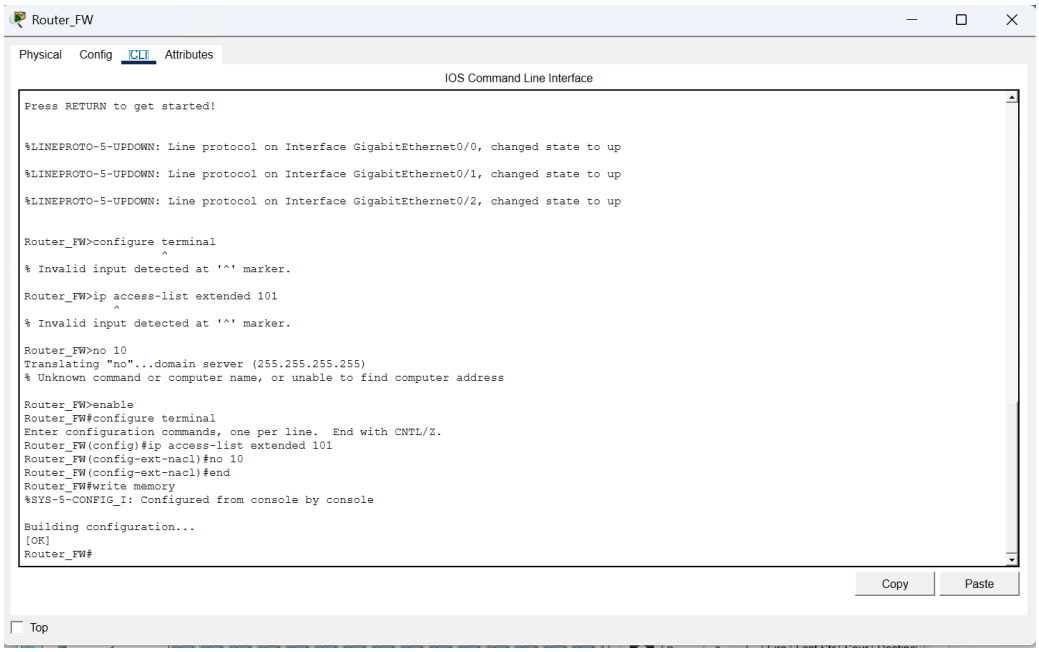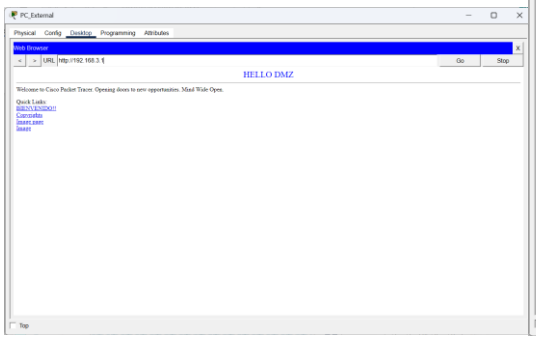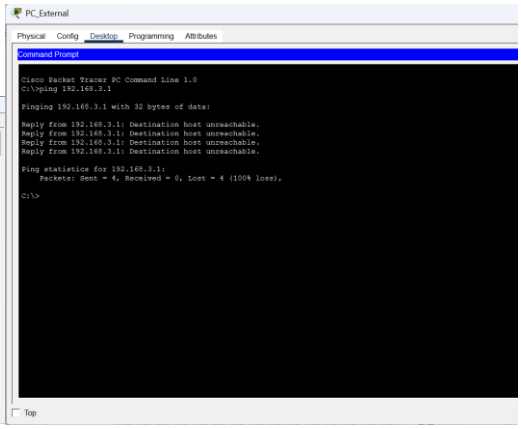
Buenas prácticas (Eliminar ICMP en producción)